Coordinate Geometry

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Distance and midpoint**

Find the distance between $A(1, 3)$ and $B(7, 11)$.

$AB = \sqrt{(7-1)^2 + (11-3)^2} = \sqrt{36+64} = \sqrt{100} = 10$.

2 **[Foundational]** • **Target: Distance and midpoint**

Find the midpoint of the segment joining $(-3, 5)$ and $(9, -1)$.

Midpoint $= \left(\frac{-3+9}{2}, \frac{5+(-1)}{2}\right) = (3, 2)$.

3 **[Foundational]** • **Target: Distance and midpoint**

The midpoint of $A(2, -3)$ and $B(x, y)$ is $(5, 1)$. Find the coordinates of B.

$\frac{2+x}{2} = 5$ gives $x = 8$. $\frac{-3+y}{2} = 1$ gives $y = 5$. So $B = (8, 5)$.

4 **[Foundational]** • **Target: Gradient and equations of lines**

Find the gradient of the line through $(2, -1)$ and $(6, 7)$.

Gradient $= \frac{7-(-1)}{6-2} = \frac{8}{4} = 2$.

5 **[Foundational]** • **Target: Parallel and perpendicular lines**

Find the equation of the line parallel to $y = 3x - 1$ passing through $(5, 0)$.

Parallel lines share the same gradient, $m = 3$: $y - 0 = 3(x - 5)$, so $y = 3x - 15$.

6 **[Foundational]** • **Target: Parallel and perpendicular lines**

Find the equation of the line perpendicular to $y = \frac{1}{3}x + 2$ passing through $(6, -1)$.

The perpendicular gradient is the negative reciprocal: $m = -3$. $y - (-1) = -3(x - 6)$, so $y = -3x + 18 - 1 = -3x + 17$.

7 **[Foundational]** • **Target: Parallel and perpendicular lines**

Determine whether the lines $y = 4x - 3$ and $4y + x = 8$ are parallel, perpendicular, or neither.

Rewrite the second line as $y = -\frac{1}{4}x + 2$, gradient $-\frac{1}{4}$. The product of the gradients is $4 \times \left(-\frac{1}{4}\right) = -1$, so the lines are perpendicular.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: The equation of a circle

Write the equation of a circle with center $(3, -2)$ and radius 6.

$(x-3)^2 + (y+2)^2 = 36$.

9 [Intermediate] • Target: The equation of a circle

Find the center and radius of $x^2 + y^2 - 4x + 6y - 12 = 0$ by completing the square.

Group: $(x^2 - 4x) + (y^2 + 6y) = 12$. Complete the square: $(x-2)^2 - 4 + (y+3)^2 - 9 = 12$, so $(x-2)^2 + (y+3)^2 = 25$. Center: $(2, -3)$. Radius: $\sqrt{25} = 5$.

10 [Intermediate] • Target: The equation of a circle

A circle has equation $(x+1)^2 + (y-4)^2 = 49$. State its center and radius.

Center: $(-1, 4)$. Radius: $\sqrt{49} = 7$.

11 [Intermediate] • Target: Lines and circles together

A circle has center $(0, 0)$ and radius 5. Determine whether the line $y = x + 8$ intersects the circle, is tangent to it, or misses it entirely, by comparing the distance from the center to the line against the radius.

Rewrite the line as $x - y + 8 = 0$. Distance from $(0, 0)$: $d = \frac{|0 - 0 + 8|}{\sqrt{1^2 + (-1)^2}} = \frac{8}{\sqrt{2}} = 4\sqrt{2} \approx 5.66$. Since $5.66 > 5$ (the distance exceeds the radius), the line misses the circle entirely.

12 [Intermediate] • Target: Lines and circles together

A circle has center $(2, 1)$ and radius 3. Determine whether the line $x + y = 10$ intersects the circle, is tangent to it, or misses it entirely.

Rewrite the line as $x + y - 10 = 0$. Distance from $(2, 1)$: $d = \frac{|2 + 1 - 10|}{\sqrt{1^2 + 1^2}} = \frac{7}{\sqrt{2}} = \frac{7\sqrt{2}}{2} \approx 4.95$. Since $4.95 > 3$, the line misses the circle entirely.

13 [Intermediate] • Target: The perpendicular bisector

Find the equation of the perpendicular bisector of the segment joining $A(1, 2)$ and $B(7, 8)$.

Midpoint: $(4, 5)$. Gradient of AB: $\frac{8-2}{7-1} = 1$. Perpendicular gradient: -1. Equation: $y - 5 = -1(x - 4)$, so $y = -x + 9$.

14 [Intermediate] • Target: Area of a triangle from coordinates

Find the area of the triangle with vertices $P(0, 0)$, $Q(8, 0)$, $R(3, 6)$.

Using the coordinate area formula: Area $= \frac{1}{2}|x_P(y_Q - y_R) + x_Q(y_R - y_P) + x_R(y_P - y_Q)| = \frac{1}{2}|0(0-6) + 8(6-0) + 3(0-0)| = \frac{1}{2}(48) = 24$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: A well-designed word problem: locating a cell tower** • **Cross-Lesson: 18.01 Binomial Distribution**

This question has two parts. Two houses are located at $A(0, 0)$ and $B(12, 0)$ (positions in kilometers). A cell tower must be built equidistant from both houses, and it must also lie on the line $y = x + 2$. (a) Find the equation of the perpendicular bisector of AB, and use it together with $y = x + 2$ to find the tower's coordinates. (b) Find the distance from the tower to each house, and confirm they are equal.

(a) Midpoint of AB: $(6, 0)$. Since AB is horizontal, its perpendicular bisector is the vertical line $x = 6$. Substituting into $y = x + 2$: $y = 8$. Tower location: $(6, 8)$. (b) Distance to A: $\sqrt{(6-0)^2 + (8-0)^2} = \sqrt{36 + 64} = \sqrt{100} = 10$ km. Distance to B: $\sqrt{(6-12)^2 + (8-0)^2} = \sqrt{36 + 64} = \sqrt{100} = 10$ km. Both distances equal 10 km, confirming the tower is equidistant from both houses.

16 **[Challenging]** • **Target: A well-designed word problem: locating a cell tower** • **Cross-Lesson: 19.01 Normal Distribution**

[Geometry & Trigonometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

17 **[Challenging]** • **Target: Distance and midpoint** • **Cross-Lesson: 2.01 Simultaneous Equations & Inequalities**

[Unit Circle & Trigonometric Identities · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

18 **[Challenging]** • **Target: Gradient and equations of lines** • **Cross-Lesson: 3.01 Graphs & Transformations**

[Conic Sections & Analytic Geometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

19 **[Challenging]** • **Target: Parallel and perpendicular lines** • **Cross-Lesson: 5.01 Trigonometry**

[Polar Coordinates, Complex Numbers & De Moivre's Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

20 **[Challenging]** • **Target: The equation of a circle** • **Cross-Lesson: 6.01 Differentiation & Integration**

[Trigonometry & Triangle Laws · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Distance and midpoint

Consider a circle with the equation $x^2 + y^2 = 25$. A point $P(x_0, y_0)$ lies on this circle. A tangent line to the circle at point P intersects the x-axis at point A and the y-axis at point B. If the area of the triangle OAB, where O is the origin $(0, 0)$, is minimized, find the value of $x_0^2 + y_0^2$.

a. **25** ✓

b. 50

c. 100

d. 125

The equation of the circle is $x^2 + y^2 = 25$. Let the point of tangency be $P(x_0, y_0)$. The equation of the tangent line to the circle at $P(x_0, y_0)$ is given by $xx_0 + yy_0 = 25$. To find the x-intercept A, we set $y = 0$: $xx_0 = 25 \Rightarrow x = \frac{25}{x_0}$. So, $A = (\frac{25}{x_0}, 0)$. To find the y-intercept B, we set $x = 0$: $yy_0 = 25 \Rightarrow y = \frac{25}{y_0}$. So, $B = (0, \frac{25}{y_0})$. The area of triangle OAB is.

22 [Foundational] • Target: Gradient and equations of lines

A sector OAB of a circle with centre O and radius 12 cm has angle $\angle AOB = \frac{\pi}{2}$ radians. The point C is the midpoint of the radius OA. The line passing through C perpendicular to OA intersects the circular arc AB at the point D. The straight line segment BC is drawn. Find the exact area, in cm^2, of the region bounded by the line segment CD, the circular arc BD, and the line segment BC.

a. $12\pi + 18\sqrt{3} - 36$ ✓

b. $24\pi + 18\sqrt{3} - 36$

c. $12\pi + 36\sqrt{3} - 36$

d. $18\pi + 18\sqrt{3} - 36$

1. **Analyze the geometry of right triangle OCD**: - C is the midpoint of OA, so $OC = \frac{1}{2}(12) = 6$ cm. - D lies on the arc of the circle, so the radius $OD = 12$ cm. - Since $CD \perp OA$, triangle OCD is a right-angled triangle at C. - $\cos(\angle COD) = \frac{OC}{OD} = \frac{6}{12} = \frac{1}{2} \Rightarrow \angle COD = \frac{\pi}{3}$ radians. 2. **Find the side length CD and the area of ΔOCD**.

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23 [Foundational] • Target: Parallel and perpendicular lines

Consider a point $P(x, y)$ that lies on the parabola defined by the equation $y = x^2 - 6x + 13$. Let F be the focus of this parabola. If the distance from P to the directrix of the parabola is minimized, what is the y-coordinate of the point P?

a. 3

b. **4** ✓

c. 5

d. 6

The given parabola is $y = x^2 - 6x + 13$. We can rewrite this equation in vertex form by completing the square: $y = (x^2 - 6x + 9) + 13 - 9$, which simplifies to $y = (x - 3)^2 + 4$. This is the equation of a parabola with vertex $(h, k) = (3, 4)$, opening upwards. The standard form of a parabola opening upwards is $(x - h)^2 = 4p(y - k)$, where (h, k) is the vertex and p is the distance from the vertex to the focus and from the vertex to the directrix. Comparing our equation $(x - 3)^2 = 1(y - 4)$ with the s.

24 [Foundational] • Target: The equation of a circle

In the triangle ABC shown, point D lies on AB and point E lies on AC such that DE is parallel to BC. Given that AD = 6, DB = 4, and the area of triangle ADE is 27 square units, find the area of triangle ABC.

a. 45

b. 60

c. **75** ✓

d. 90

Since DE is parallel to BC, by the property of similar triangles, triangle ADE is similar to triangle ABC. The ratio of corresponding sides is $\frac{AD}{AB} = \frac{AD}{AD + DB} = \frac{6}{6 + 4} = \frac{6}{10} = \frac{3}{5}$. For similar triangles, the ratio of their areas is the square of the ratio of their corresponding sides. Therefore, $\frac{\text{Area}(ADE)}{\text{Area}(ABC)} = \left(\frac{AD}{AB}\right)^2 = \left(\frac{3}{5}\right)^2 = \frac{9}{25}$. We are given that the area of triangle ADE is 27 square.

25 [Foundational] • Target: Lines and circles together

The locus $|z - (-6 + i)| = |z - (2 - 2i)|$ is the perpendicular bisector of the segment joining the points representing z_1 and z_2. Find the x-coordinate of the midpoint of this segment.

a. $-\frac{1}{2}$

b. 2

c. **-2** ✓

d. -6

Midpoint x-coordinate: $\frac{-6 + 2}{2} = -2$.

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26 [Foundational] • Target: The perpendicular bisector

A circle is centered at the origin and passes through the point $P(a, 4)$ in the first quadrant. The equation of a tangent line to the circle at point P is given by $3x + 4y = k$. Find the value of k.

a. **25** ✓ **b.** 30

c. 35 **d.** 40

The equation of a circle centered at the origin is $x^2 + y^2 = r^2$. Since point $P(a, 4)$ lies on the circle, we have $a^2 + 4^2 = r^2$, so $a^2 + 16 = r^2$. The radius from the origin to $P(a, 4)$ is perpendicular to the tangent line at P. The slope of the radius is $m_{radius} = \frac{4-0}{a-0} = \frac{4}{a}$. The given tangent line is $3x + 4y = k$, which can be rewritten in slope-intercept form as $y = -\frac{3}{4}x + \frac{k}{4}$. The slope of the tangent line is $m_{tangent} = -\frac{3}{4}$. Since the rad.

27 [Foundational] • Target: Area of a triangle from coordinates

A sector of a circle with radius 6 cm has area 27 cm^2. Find the angle of the sector, in radians.

a. **1.5** ✓ **b.** 3

c. 0.75 **d.** 2

$27 = \frac{1}{2}(6^2)\theta = 18\theta \Rightarrow \theta = 1.5$ rad.

28 [Foundational] • Target: A well-designed word problem: locating a cell tower

The line l_1 has vector equation $\mathbf{r} = (1; 5; 0) + \lambda(1; -2; 2)$ and the line l_2 has vector equation $\mathbf{r} = (1; p; 2) + \mu(2; 3; q)$ where p and q are constants, and $\lambda, \mu \in \mathbb{R}$. Given that the lines l_1 and l_2 are perpendicular and intersect at a common point, find the value of p.

a. -2 ✓ **b.** 2

c. -4 **d.** 4

We solve the problem in three steps: **Step 1: Determine q using perpendicularity** The direction vectors of l_1 and l_2 are $\mathbf{d}_1 = (1; -2; 2)$ and $\mathbf{d}_2 = (2; 3; q)$. Since $l_1 \perp l_2$, their scalar (dot) product must equal zero: $\mathbf{d}_1 \cdot \mathbf{d}_2 = (1)(2) + (-2)(3) + (2)(q) = 0\ 2 - 6 + 2q = 0 \Rightarrow 2q = 4 \Rightarrow q = 2.$ **Step 2: Find the parameters λ and μ at the intersection** Equating the coordinates of l_1 and l_2.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 **[Intermediate]** • **Target: Distance and midpoint**

In triangle PQR, angle PQR = 90 degrees. S is a point on PR such that QS is perpendicular to PR. The length of PQ is 15 cm and the length of QR is 20 cm. Let the length of PS be x cm and the length of SR be y cm. Find the value of x.

a. 10 cm

b. **12 cm ✓**

c. 15 cm

d. 9 cm

First, we find the length of the hypotenuse PR using the Pythagorean theorem in triangle PQR: $PR^2 = PQ^2 + QR^2$. $PR^2 = 15^2 + 20^2 = 225 + 400 = 625$. So, $PR = \sqrt{625} = 25$ cm. Triangle PQR is similar to triangle PSQ (by AA similarity, as both share angle P and have a right angle). Therefore, the ratio of corresponding sides is equal: $\frac{PS}{PQ} = \frac{PQ}{PR}$. Substituting the given values: $\frac{x}{15} = \frac{15}{25}$. Solving for x: $x = 15 \times \frac{15}{25} = 15 \times \frac{3}{5} = 9$ cm..

30 **[Intermediate]** • **Target: Gradient and equations of lines**

Consider a point $P(\cos\theta, \sin\theta)$ on the unit circle in the Cartesian plane. If the distance from the origin to the point $Q(x, y)$ is 5, and the angle between the positive x-axis and the line segment OQ is α, such that $\cos\theta = \frac{3}{5}$ and $\sin\theta = \frac{4}{5}$. If $x = \cos\alpha$ and $y = \sin\alpha$, find the maximum possible value of the product of the x-coordinates of P and Q, i.e., $\cos\theta \cdot \cos\alpha$.

a. 4/5

b. 3/5

c. **1 ✓**

d. 5/3

The point P is on the unit circle, so its coordinates are $(\cos\theta, \sin\theta)$. We are given $\cos\theta = \frac{3}{5}$ and $\sin\theta = \frac{4}{5}$. The point Q has coordinates (x, y) and its distance from the origin is 5. The angle between the positive x-axis and OQ is α. Thus, $x = 5\cos\alpha$ and $y = 5\sin\alpha$. We are asked to find the maximum possible value of the product of the x-coordinates of P and Q, which is $\cos\theta \cdot x$. Substituting the given values, we h.

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31 [Intermediate] • Target: Parallel and perpendicular lines

Consider a circle with its center at the origin $(0, 0)$ and a radius of 5 units. A chord of this circle is perpendicular to the x-axis and intersects the x-axis at the point $(3, 0)$. The endpoints of this chord on the circle are A and B. Determine the length of the chord AB.

a. 6

b. **8** ✓

c. 10

d. 4

The equation of the circle centered at the origin with radius 5 is $x^2 + y^2 = 5^2 = 25$. The chord is perpendicular to the x-axis and intersects it at $(3, 0)$. This means the x-coordinate of the endpoints of the chord, A and B, is 3. Substitute $x = 3$ into the circle's equation to find the corresponding y-coordinates: $3^2 + y^2 = 25$ $9 + y^2 = 25$ $y^2 = 25 - 9$ $y^2 = 16$ $y = \pm 4$ So the endpoints of the chord are $A = (3, 4)$ and $B = (3, -4)$. The length of the chord AB is the distance between these t.

32 [Intermediate] • Target: The equation of a circle

A circle has its center at the origin $(0, 0)$ and passes through the point $P(3, y)$. If the equation of the circle is $x^2 + y^2 = 25$, what is the positive value of y?

a. 3

b. **4** ✓

c. 5

d. 16

The equation of a circle centered at the origin with radius r is given by $x^2 + y^2 = r^2$. In this problem, the equation of the circle is given as $x^2 + y^2 = 25$. This implies that the radius squared, r^2, is equal to 25. The circle passes through the point $P(3, y)$. Substituting the coordinates of point P into the equation of the circle, we get: $3^2 + y^2 = 25$. This simplifies to $9 + y^2 = 25$. To find the value of y^2, we subtract 9 from both sides: $y^2 = 25 - 9$, which gives $y^2 = 16$. Taking t.

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33 [Intermediate] • Target: Lines and circles together

In the diagram, points A, B, and C form a triangle. Point D lies on AC such that BD is the angle bisector of angle ABC. Given that AB = 10 cm, AC = 15 cm, and BC = 12 cm, and that AD = 6 cm, find the length of DC.

a. 8 cm

b. **7.2 cm ✓**

c. 9 cm

d. 6 cm

According to the Angle Bisector Theorem, an angle bisector of a triangle divides the opposite side into two segments that are proportional to the other two sides of the triangle. In triangle ABC, BD is the angle bisector of angle ABC. Therefore, we have the proportion: $\frac{AD}{DC} = \frac{AB}{BC}$ We are given that AB = 10 cm, AC = 15 cm, and BC = 12 cm. We are also given that AD = 6 cm. Let DC = x cm. Substituting the given values into the proportion: $\frac{6}{x} = \frac{10}{12}$ To solve for x, we can cross-m.

34 [Intermediate] • Target: The perpendicular bisector

In the diagram, triangle ABC has angle ABC equal to 90 degrees. Point D lies on AC such that BD is perpendicular to AC. If AD = 4 cm and DC = 9 cm, find the length of BD.

a. 5 cm

b. **6 cm ✓**

c. 7 cm

d. 3 cm

In a right-angled triangle, when an altitude is drawn from the right angle to the hypotenuse, it creates two smaller triangles that are similar to the original triangle and to each other. This leads to the geometric mean theorem. The length of the altitude (BD) is the geometric mean of the two segments it divides the hypotenuse into (AD and DC). Therefore, $BD^2 = AD \times DC$. Given AD = 4 cm and DC = 9 cm, $BD^2 = 4 \times 9$ $BD^2 = 36$ $BD = \sqrt{36}$ $BD = 6$ cm.

35 [Intermediate] • Target: Area of a triangle from coordinates

In the given triangle ABC, AB = 10, AC = 15, and BD = 6. D is a point on BC such that AD is the angle bisector of angle BAC. Find the length of DC.

a. **9 ✓**

b. 10

c. 15

d. 22.5

By the Angle Bisector Theorem, the ratio of the sides adjacent to the angle bisector is equal to the ratio of the segments created by the bisector on the opposite side. Therefore, $\frac{AB}{AC} = \frac{BD}{DC}$. Substituting the given values, we get $\frac{10}{15} = \frac{6}{DC}$. Simplifying the left side, $\frac{2}{3} = \frac{6}{DC}$. Cross-multiplying, $2 \times DC = 3 \times 6$. $2 \times DC = 18$. Dividing by 2, $DC = \frac{18}{2} = 9$.

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36 [Intermediate] • Target: A well-designed word problem: locating a cell tower

In the triangle ABC, AB = 10, AC = 15, BD = 6, and D is a point on BC such that AD is the angle bisector of angle BAC. Find the length of DC.

a. 7

b. **9** ✓

c. 10

d. 12

According to the Angle Bisector Theorem, the ratio of the sides adjacent to the angle bisector is equal to the ratio of the segments it divides the opposite side into. Therefore, $\frac{AB}{AC} = \frac{BD}{DC}$. Substituting the given values, we have $\frac{10}{15} = \frac{6}{x}$, where x is the length of DC. Solving for x: $10x = 15 \times 6$ $10x = 90$ $x = \frac{90}{10}$ $x = 9$.

37 [Intermediate] • Target: Distance and midpoint

A circle is centered at the origin $(0, 0)$ and passes through the point $(3, 4)$. What is the equation of this circle?

a. $x^2 + y^2 = 7$

b. $x^2 + y^2 = 9$

c. $x^2 + y^2 = 16$

d. $x^2 + y^2 = 25$ ✓

The standard equation of a circle centered at the origin $(0, 0)$ with radius r is $x^2 + y^2 = r^2$. Since the circle passes through the point $(3, 4)$, we can substitute these coordinates into the equation to find r^2. $3^2 + 4^2 = r^2$. This gives $9 + 16 = r^2$, so $r^2 = 25$. Therefore, the equation of the circle is $x^2 + y^2 = 25$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Gradient and equations of lines • Cross-Lesson: 3.01 Graphs & Transformations

What is the radius of the circle with the equation $(x - 3)^2 + (y + 5)^2 = 36$?

a. **6** ✓

b. 36

c. 3

d. 5

The standard equation of a circle is $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) is the center of the circle and r is the radius. In the given equation $(x - 3)^2 + (y + 5)^2 = 36$, we can see that $r^2 = 36$. To find the radius r, we take the square root of both sides: $r = \sqrt{36} = 6$. Therefore, the radius of the circle is 6.

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39 [Challenging] • Target: Parallel and perpendicular lines • Cross-Lesson: 5.01 Trigonometry

The area of a sector is 3 times its arc length for a circle of radius r. Find r.

a. **6** ✓

b. 3

c. 12

d. 9

Area $= 12r^2$, arc length $= r\theta$. Setting area $= 3 \times$ arc length: $12r^2 = 3r12r = 3r = 6$ (dividing both sides by $r\theta eq 0$).

40 [Challenging] • Target: The equation of a circle • Cross-Lesson: 6.01 Differentiation & Integration

Points $A(1, 0)$ and $B(7, -6)$ are given. Find the gradient of the perpendicular bisector of $[AB]$.

a. -1

b. 0

c. 2

d. **1** ✓

Working through this modelling step: Gradient of AB: $\frac{-6}{6}$. Perpendicular gradient $= -\frac{1}{\text{grad}_{AB}} = 1$.

41 [Challenging] • Target: Lines and circles together • Cross-Lesson: 7.01 Statistics & Probability

A sector of a circle has radius 9 cm and angle 45°. Find the exact area of the sector.

a. $\frac{81\pi}{8}$ ✓

b. $\frac{49\pi}{8}$

c. 8π

d. $\frac{9\pi}{2}$

$A = \frac{1}{2}r^2\theta = \frac{1}{2}(9)^2\left(\frac{1\pi}{4}\right) = \frac{81\pi}{8}$ cm^2.

42 [Challenging] • Target: The perpendicular bisector • Cross-Lesson: 8.01 Functions: Composite & Inverse

The polar equation $r = 2\cos\theta$ represents a circle. What is its radius?

a. 2

b. 3

c. 5

d. **1** ✓

$r = 2a\cos\theta$ is the circle $(x - a)^2 + y^2 = a^2$ with $2a = 2 \Rightarrow a = 1$; radius $= 1$.

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43 [Challenging] • Target: Area of a triangle from coordinates • Cross-Lesson: 9.01 Circular Measure

In triangle ABC, $a = 6$, $b = 8$, and $C = 90^\circ$. Find the exact area of the triangle.

a. 48

b. $24\sqrt{2}$

c. **24** ✓

d. $24\sqrt{3}$

Area $= \frac{1}{2}ab\sin C = \frac{1}{2}(6)(8)\sin(90^\circ) = 24$.

44 [Challenging] • Target: A well-designed word problem: locating a cell tower • Cross-Lesson: 1.01 Indices, Surds & Quadratics

Consider the unit circle shown, where a point $P(x, y)$ lies on the circle in the first quadrant. A ray from the origin through P forms an angle θ with the positive x-axis. If the y-coordinate of P is $\frac{\sqrt{3}}{2}$ and $\cos\theta = \frac{1}{2}$, find the value of x.

a. $\frac{\sqrt{3}}{2}$

b. $\frac{1}{2}$ ✓

c. 1

d. $\frac{\sqrt{2}}{2}$

The equation of a unit circle centered at the origin is $x^2 + y^2 = 1$. We are given that the y-coordinate of point P is $y = \frac{\sqrt{3}}{2}$. Substituting this into the equation of the unit circle: $x^2 + (\frac{\sqrt{3}}{2})^2 = 1$ $x^2 + \frac{3}{4} = 1$ $x^2 = 1 - \frac{3}{4}$ $x^2 = \frac{1}{4}$ Since point P lies in the first quadrant, its x-coordinate must be positive. Therefore, $x = \sqrt{\frac{1}{4}} = \frac{1}{2}$. Alternatively, we are given that the angle θ formed by the ray from the or.

45 [Challenging] • Target: Distance and midpoint • Cross-Lesson: 10.01 Series & Binomial Theorem

In the right-angled triangle ABC shown below, the length of side AB is 10 units and the length of side BC is 6 units. The angle ACB is 90 degrees. Find the length of side AC.

a. 10

b. 6

c. **8** ✓

d. 16

The triangle ABC is a right-angled triangle with the right angle at C. We can use the Pythagorean theorem, which states that in a right-angled triangle, the square of the hypotenuse (the side opposite the right angle) is equal to the sum of the squares of the other two sides. In this case, AB is the hypotenuse. So, we have: $AB^2 = AC^2 + BC^2$. We are given $AB = 10$ and $BC = 6$. We need to find AC. Substituting the given values into the equation: $10^2 = AC^2 + 6^2$ $100 = AC^2 + 36$ $AC^2 = 100 - 36$.

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